



Current and Emerging Technology Trends Among Developers: A Data Analysis of the 2019 Stack Overflow Survey

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OUTLINE



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- Introduction
- Methodology
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 - Visualization – Charts
 - Dashboard
- Discussion
 - Findings & Implications
- Conclusion
- Appendix



EXECUTIVE SUMMARY



- Most Used Languages: JavaScript leads; Python growing fast.
- Most Used Languages: JavaScript and SQL lead
 - Web and Full-stack: Most common roles
 - DevOps: Gaining popularity
 - Data Roles: Driven by Python and R
- Platforms: AWS top the list of the most platform to work with.
- Databases Trends: PostgreSQL remains the most used database



INTRODUCTION



- This presentation is based on Stack Overflow's Developer Survey: the largest global survey of nearly 90,000 developers
- Analyzes current and desired technologies used by developers
- Offers demographic insights to guide hiring and training strategies
- Supports data-driven decisions for tech investment and developer upskilling
- Identifies global developer trends and predicts future directions



METHODOLOGY



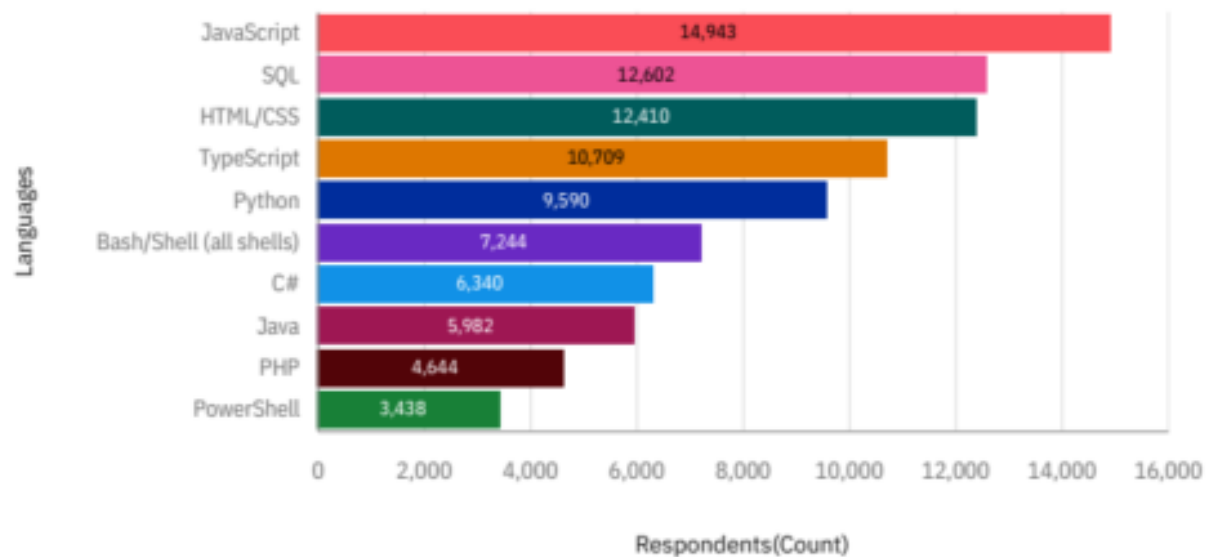
- Data Collection Stack
 - Overflow Developer Survey data extraction
 - GitHub Jobs API
 - Dataset Techniques: Web Scraping, APIs (using Python requests)
- Data Preparation :
 - Data cleaning: Remove duplicates, handle missing values, normalize data
 - Wrangling: Reshape and merge datasets for analysis
- Exploratory Data Analysis (EDA)
 - Analyze distributions, detect outliers, explore correlations
 - Tools: Python (Pandas, Matplotlib, Seaborn)Data
- Data Visualization
 - Visualize distributions, relationships, and comparisons
 - Create interactive dashboards to highlight trends and insights
- Dashboard Development
 - IBM Cognos Dashboard Embedded (CDE) and Google Looker Studio used for final visualization and storytelling



PROGRAMMING LANGUAGE TRENDS

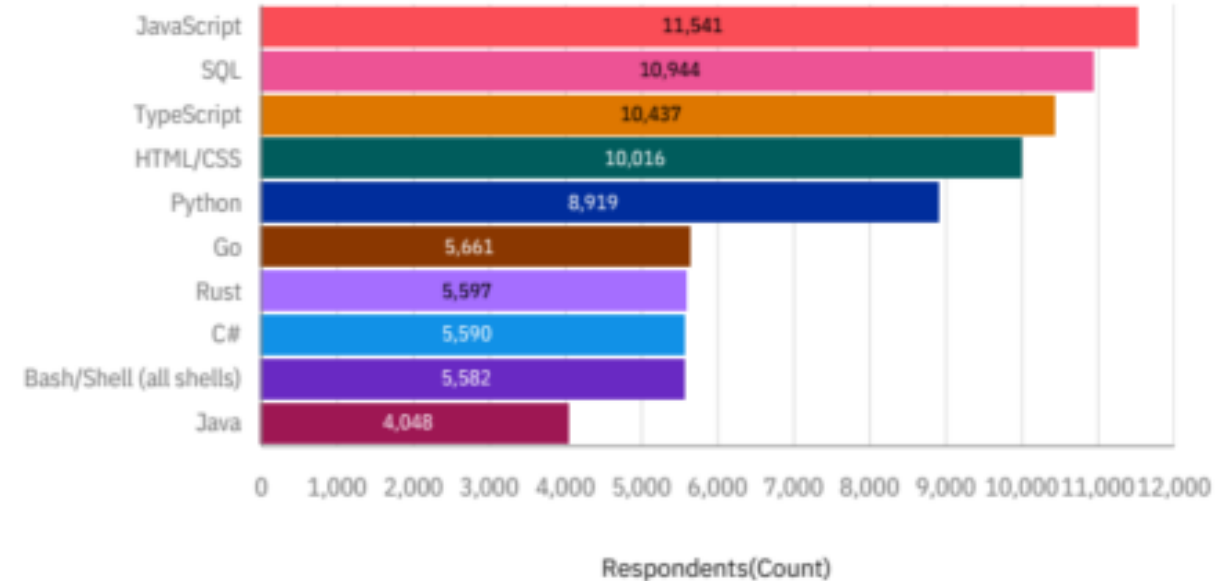
Current Year

Top 10 Languages Worked with



Next Year

Top 10 Languages Desired to Work With



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

- Web Technologies Dominate: JavaScript, TypeScript, HTML, and CSS continue to lead in usage.
- Growth in Systems Languages: Go and Rust are gaining traction.
- SQL and Python Remain Consistent: These foundational languages maintain stable interest.
- Enterprise Languages Hold Steady: C# and Java continue to see reliable demand.
- Decline in Legacy Technologies: PHP and PowerShell experience reduced interest.

Implications

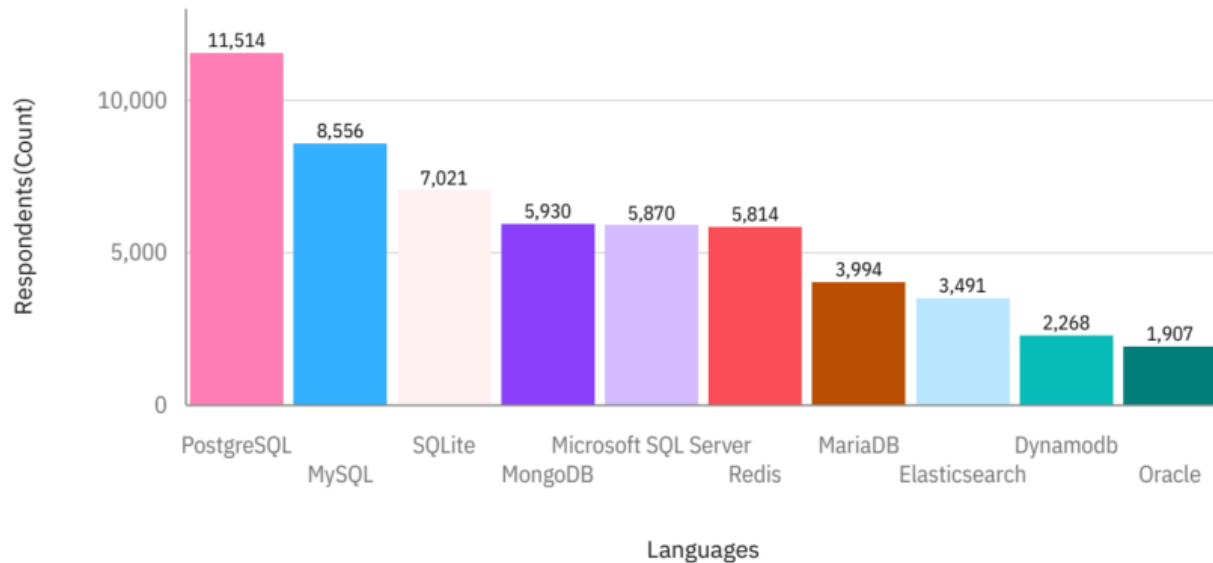
- The slight decline in interest for web technologies may reflect market saturation and maturity in the field.
- Python's sustained popularity is likely driven by its central role in AI and machine learning applications
- The decline in PHP and PowerShell suggests a shift away from older, legacy systems toward modern alternatives.
- PHP and PowerShell are no longer the languages of interest
- Emerging languages like Rust and Go present valuable opportunities for differentiation and relevance in next-gen computing fields.



DATABASE TRENDS

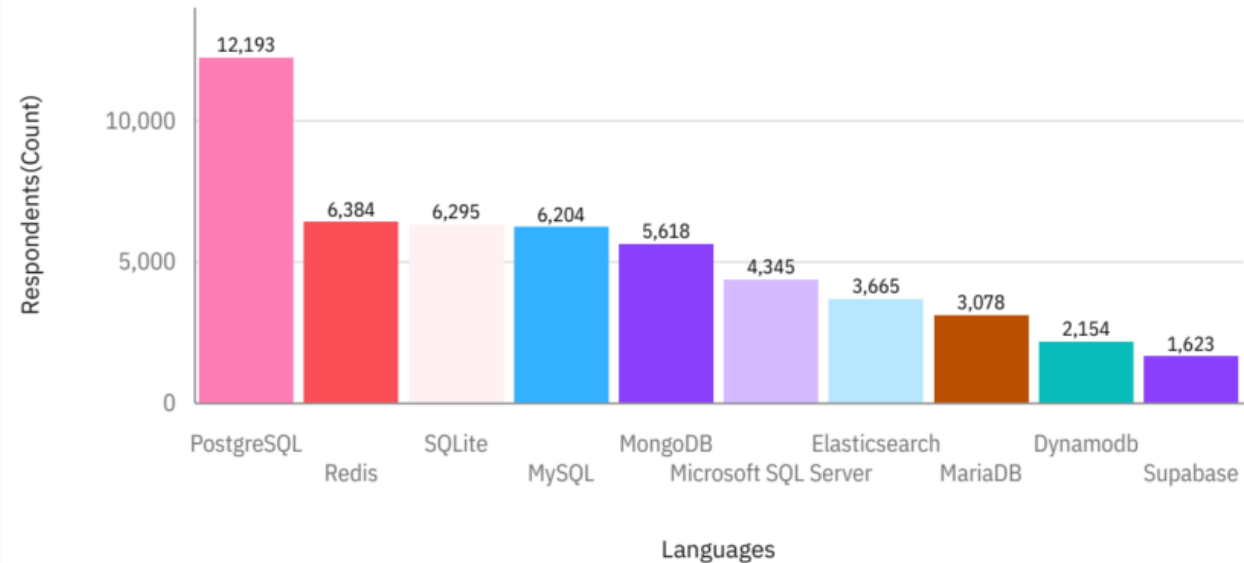
Current Year

Top 10 Databases Worked with



Next Year

Top 10 Databases Desired to Work with



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL stands out as the clear leader among database technologies.
- Redis emerges as a fast-rising contender in the database landscape.
- Interest in Oracle has declined noticeably

Implications

- The strong preference for PostgreSQL suggests that organizations value its scalability, open-source nature, and extensibility.
- The rise of Redis may reflect the growing demand for real-time data processing in modern applications.
- The decline in Oracle's popularity indicates a broader industry shift away from traditional proprietary systems toward more flexible and cost-effective solutions.



DASHBOARD



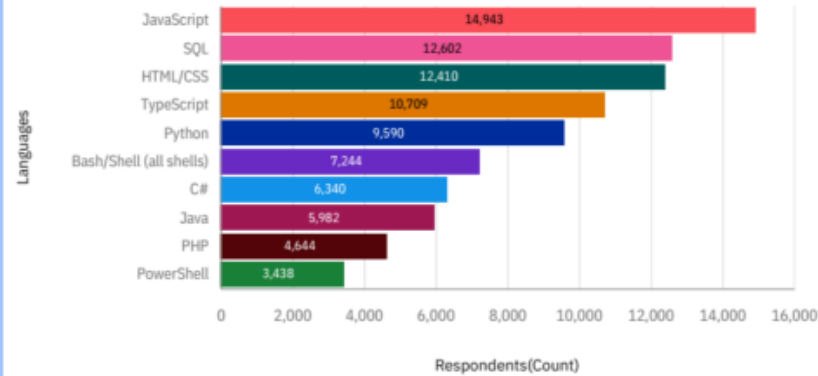
Dashboard Presentation



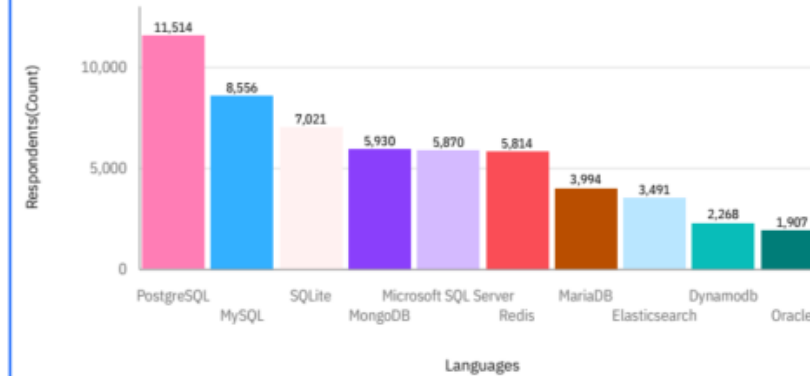
DASHBOARD TAB 1

Current Technology Usage

Languages Weorked With



Databases Worked With



Platforms Worked With



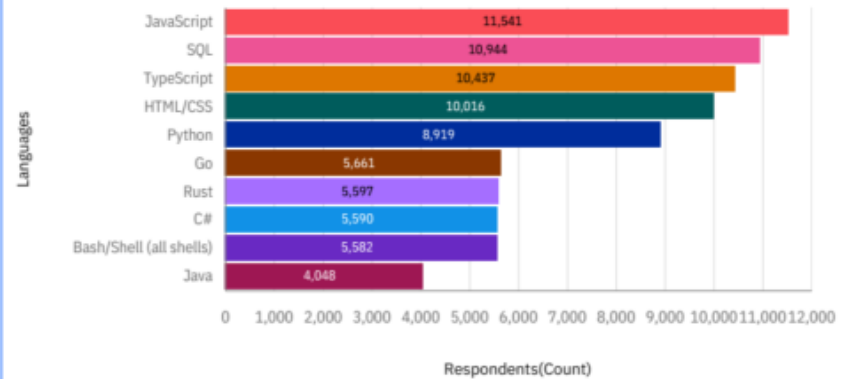
Webframes Worked With



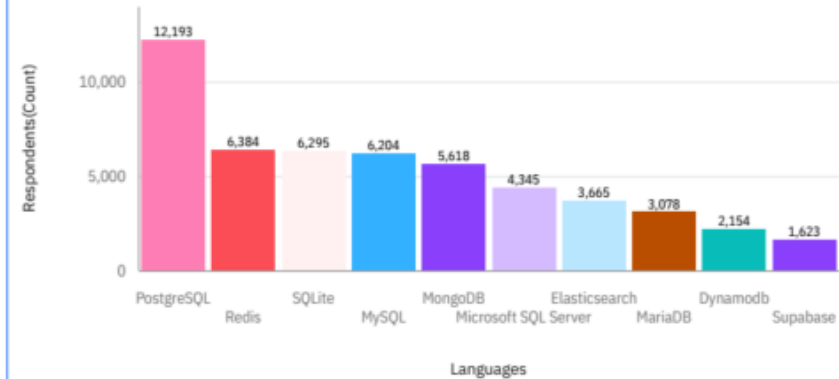
DASHBOARD TAB 2

Future Technology Trend

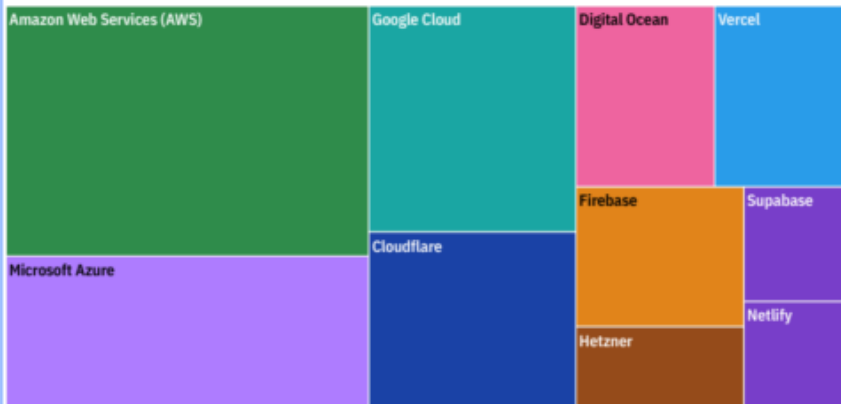
Languages Desired to Work With



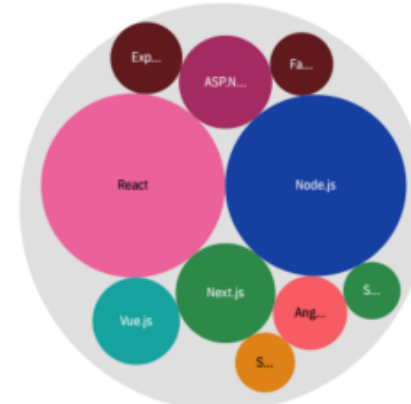
Databases Desired to Work With



Platforms Desired to Work With



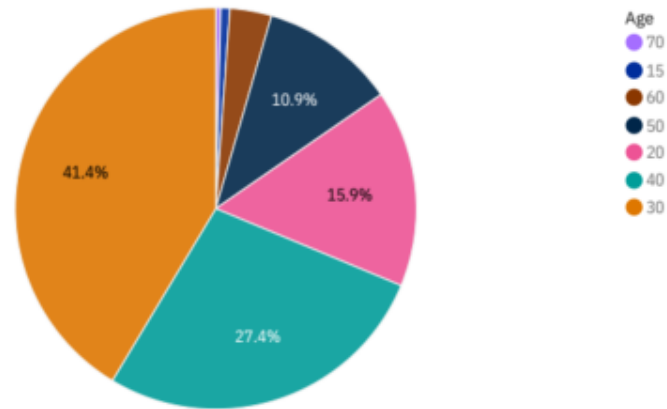
Webframes Desired to Work With



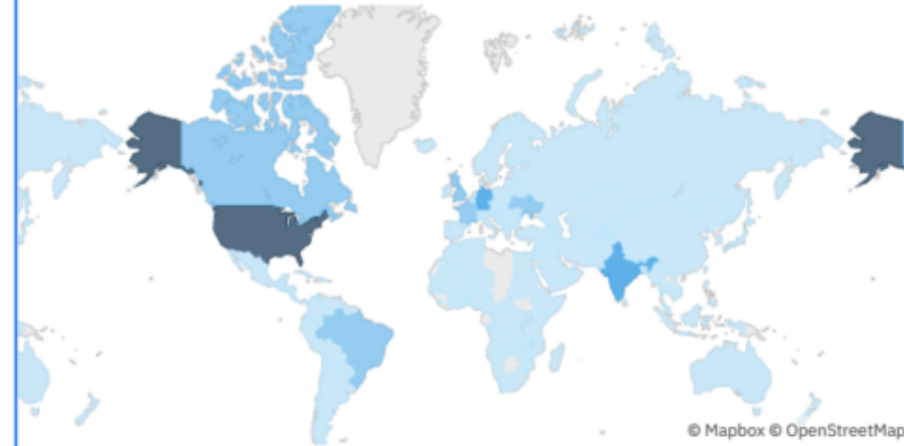
DASHBOARD TAB 3

Demographics

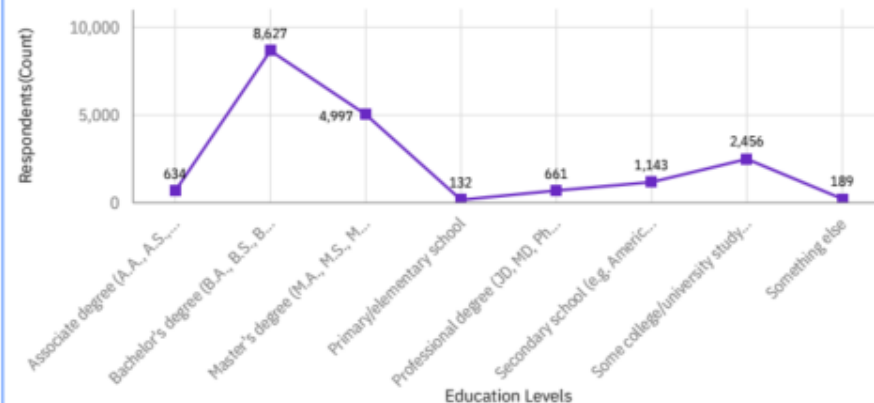
Pie Chart of Respondent distribution by Age



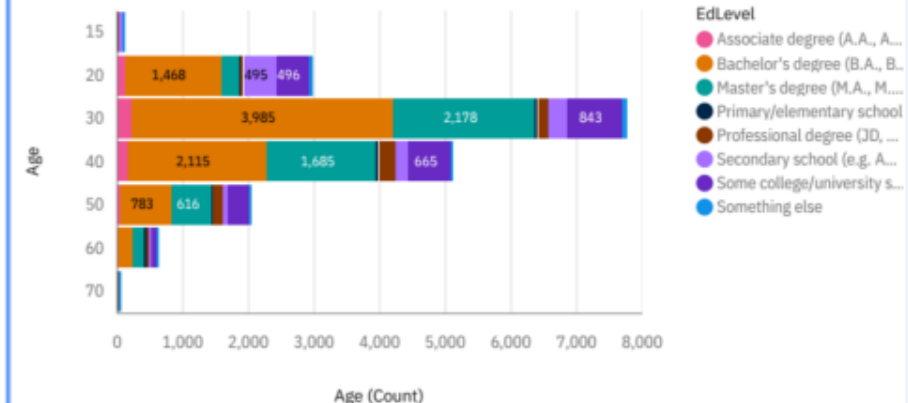
Respondent Count by Country



Respondent Count by Country



Respondent Count by Age, classified by Education Level



DISCUSSION



- **Web development languages (Javascript and HTML/CSS) and SQL** are the most commonly used technologies among job postings.
- **PostgreSQL** remains the top database choice, reflecting its reliability and strong industry preference.
- **Redis** is gaining momentum as many professionals are transitioning to it from other databases.
- **AWS** continues to be the dominant cloud platform, both in usage and in demand.
- **Node.js and React** lead among web frameworks, maintaining steady popularity across postings.



OVERALL FINDINGS & IMPLICATIONS

Findings

- Web development languages continue to grow in demand
- Open-source RDBMS dominates
- Redis is the breakout star
- Developed countries have higher numbers of respondents
- Enterprise DBs (Oracle, MS SQL) stagnate

Implications

- Continue to train workers on web development languages
- The scalability of PostgreSQL makes it the top choice
- Developing countries have limited numbers of developers



CONCLUSION



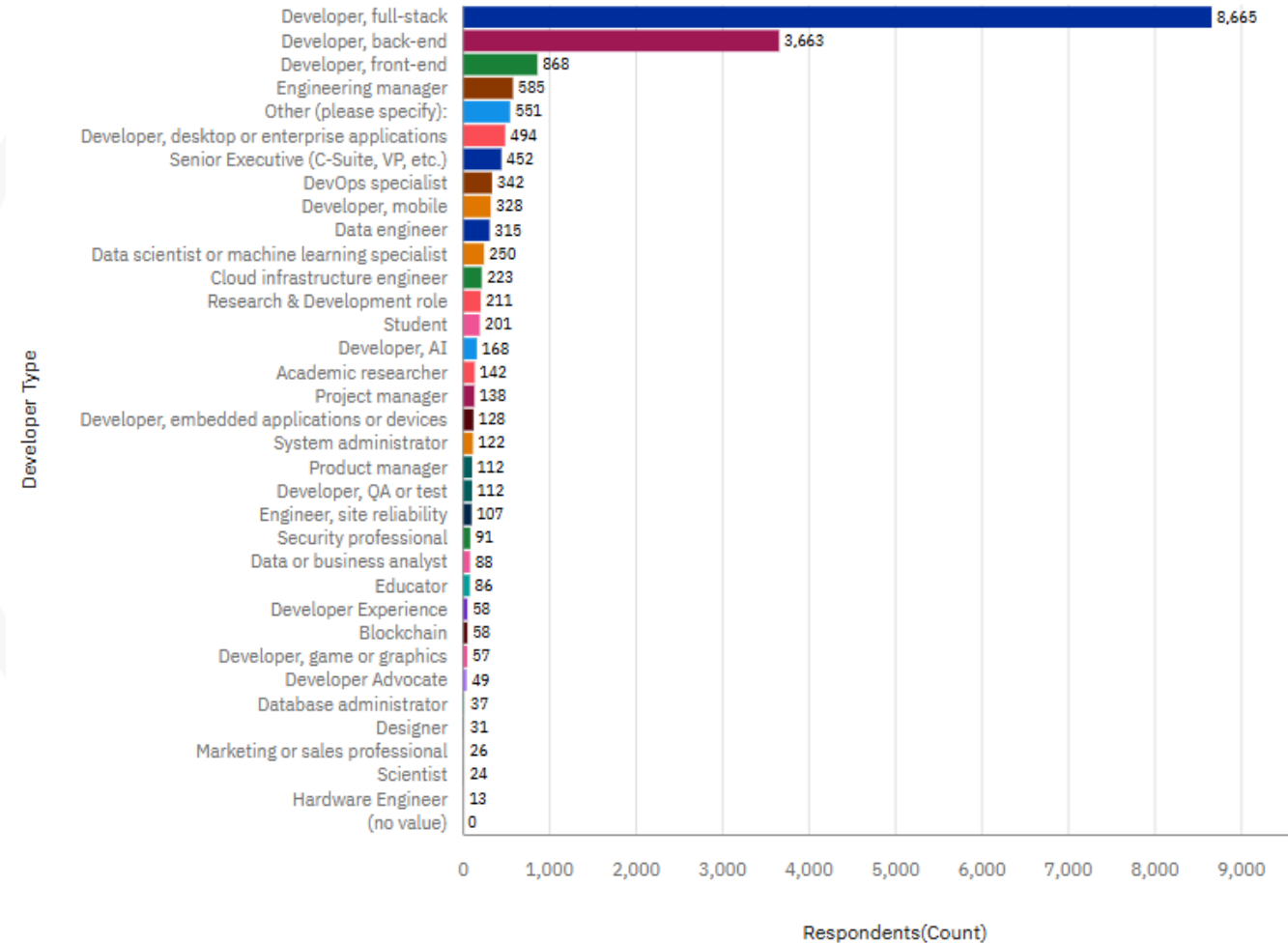
- **Investing in technology training** is essential, especially in developing countries, to bridge skill gaps and foster innovation
- **Continuous upskilling in web development languages** is vital to maintain relevance in a rapidly evolving tech landscape.
- Developers are encouraged to **prioritize in-demand technologies** such as PostgreSQL, Redis, Go, and Rust to stay competitive.
- The dominance of **open-source RDBMS** highlights the need to promote and support open-source tools in developer communities.



APPENDIX



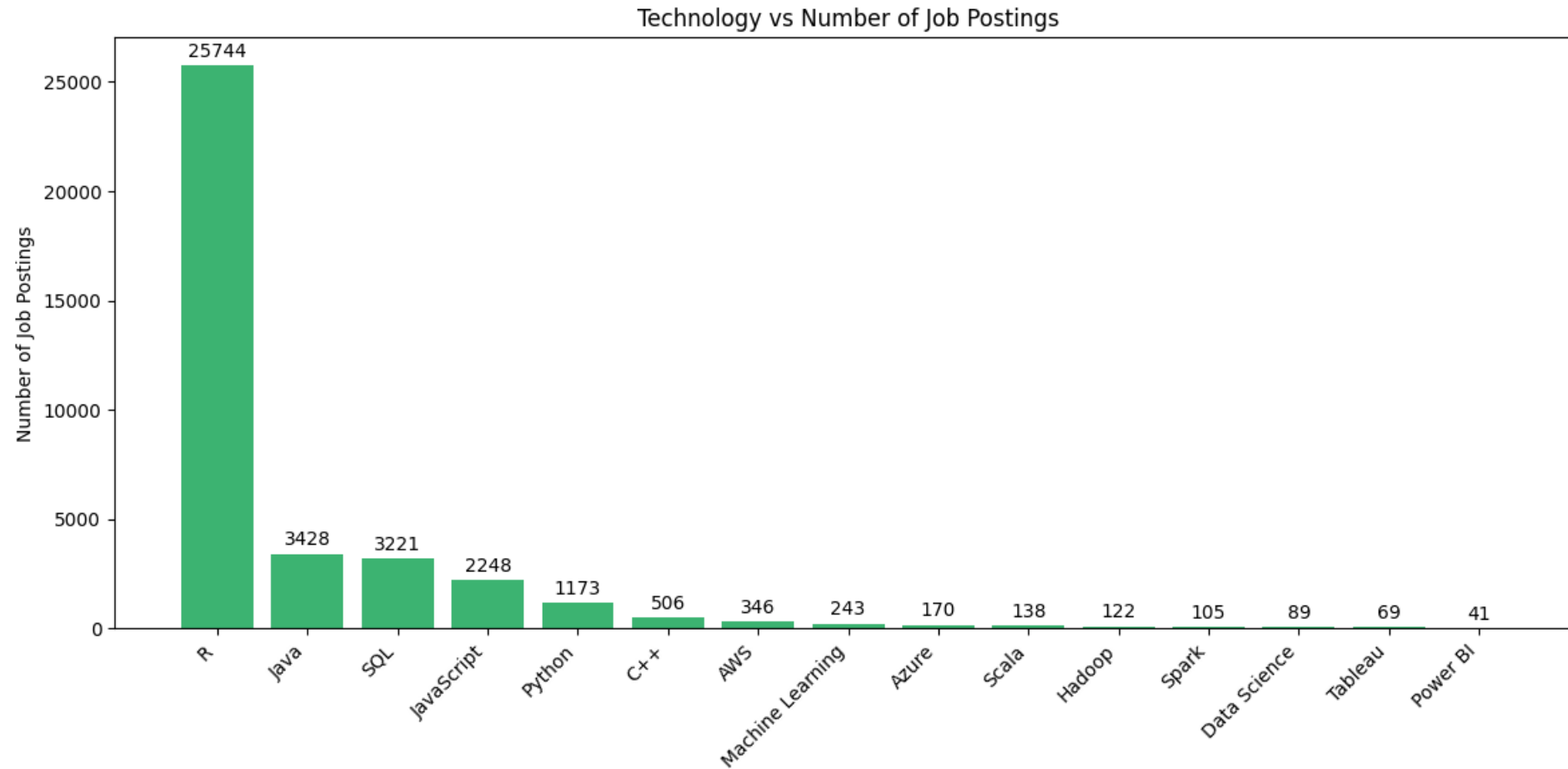
Developer Type



Full Stack developers are highly sought after



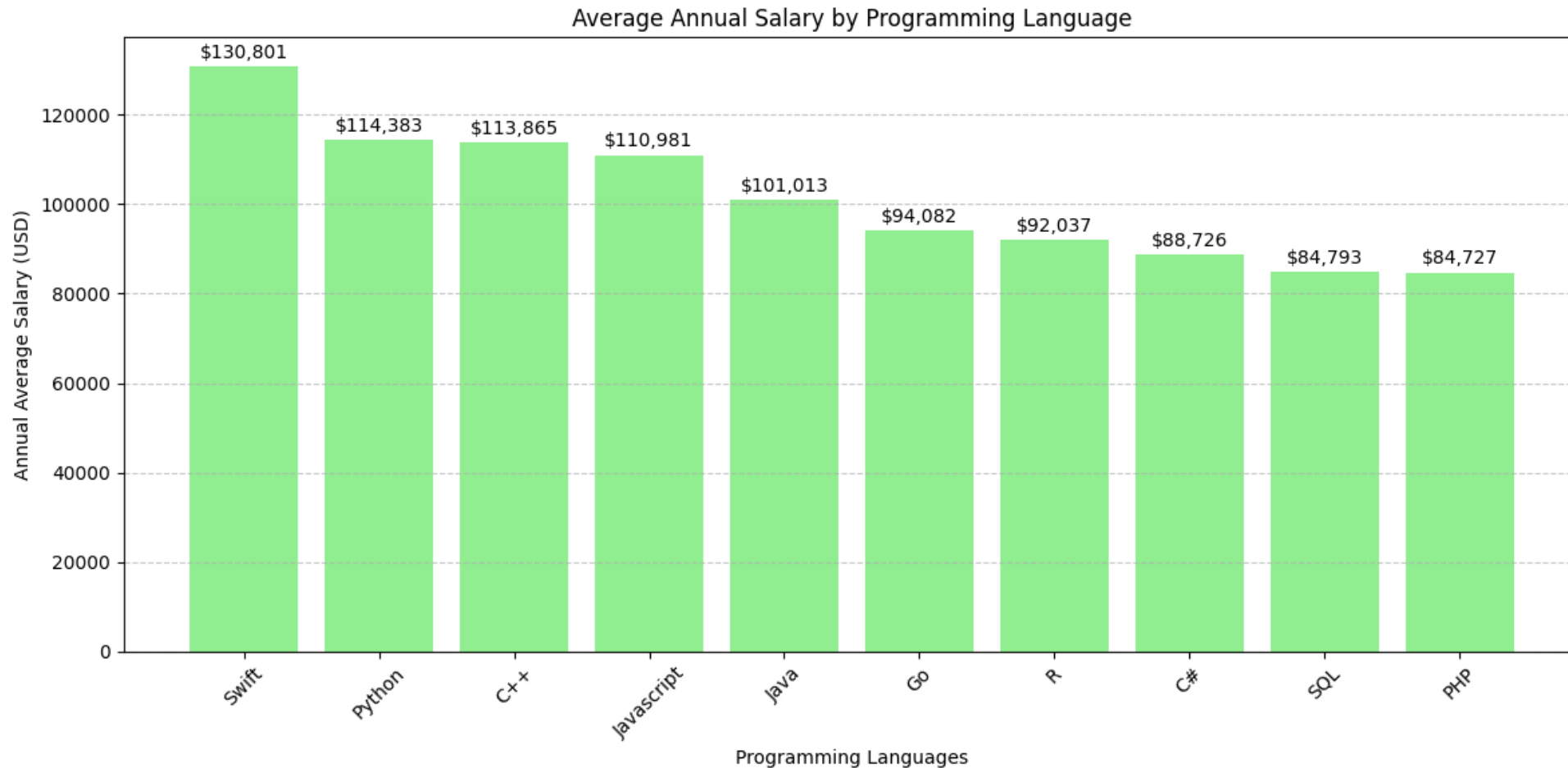
JOB POSTINGS



Job posting
by
Technology



POPULAR LANGUAGES



Popular languages and their average annual salaries. Data extracted from GitHub using Web scraping python tool: BeautifulSoup.

